

SYLLABUS: BASIC PHYSICS

COURSE CONTENT

Block 1: Basic physics

Unit 1: States of matter

Unit 2: Changes of state

Unit 3: Gas behavior under changing conditions

Gas laws: Boyles / Charles / Gay Lussais, Daltons laws & application

Unit 4: Fluid dynamics

Concepts of pressure, volume, flow, Temperature, Humidity Measurements – units & devices Introduction to Medical term describing normal & abnormal process

Block 2: Medical Gases:

Unit 1: Characteristics of Medical gases

Unit 2: Storage of medical gases

Section 1: Cylinders, Liquid gas storage, oxygen concentrator

Unit 3: Distribution of regulation of medical gases

Section 1: Piped distribution system

Reference Books

1. Davis P: Basic Physics and Measurement Anesthesia.
2. Thayalan K: Bio Medical Physics for Nurses.
3. Kacmarek R. M: Egan's Fundamentals of Respiratory Care. Elsevier(2013)